



Southeast University

Introduction to Software Engineering

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Outline

Part 0. Overview

Part 1. Software Processes

Part 2. Modeling

❖ **Chap 7. Understanding Requirements**

❖ **Chap 8-10. Requirements Modeling**

❖ **Chap 11. Design Concepts**

❖ **Chap 12-14. Design Methods**

Part 3. Quality Assurance

Part 4. Software Project Management

Chap 14. User Interface Design

- ❖ **1. Golden Rules for UI Design**
- ❖ **2. User Interface Design Models**
- ❖ **3. User Interface Design Process**
 - **Interface Analysis**
 - User Analysis
 - Task Analysis and Modeling
 - Analysis of Display Content
 - the environment
 - **Interface Design Steps & Design Issues**
 - **Design Evaluation Cycle**

Background

❖ 在上海发布的《关于全面推进上海城市数字化转型的意见》中，提出

- “构建充满活力的数字生活服务生态，形成人人享有更具品质、更加美好的数字生活新范式。”
- “倡导各类公共服务‘**数字无障碍**’，面向老年人和残障人士推进相关服务的适应性改造，创造无处不在、优质普惠的数字生活新图景。”

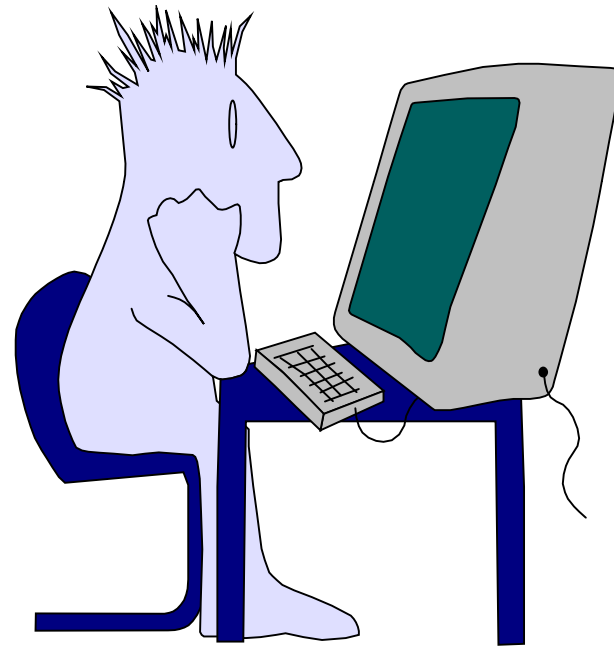


Interface Design

Easy to learn?

Easy to use?

Easy to understand?



Interface Design

Typical Design Errors

lack of consistency
too much memorization
no guidance / help
no context sensitivity
poor response
Arcane/unfriendly



Golden Rules

- ❖ **Place the user in control**
- ❖ **Reduce the user's memory load**
- ❖ **Make the interface consistent**

1. 用户接口设计的黄金规则

- 用户控制操作
- 减少用户记忆负担
- 保存界面一致

Place the User in Control

- ❖ Define interaction modes in a way that **does not force a user** into unnecessary or undesired actions.
- ❖ Provide **for flexible interaction**.
- ❖ **Allow** user interaction to be **interruptible and undoable**.
- ❖ Streamline interaction as skill levels advance and allow the interaction to be customized.
- ❖ **Hide technical internals** from the casual user.
- ❖ Design for **direct interaction with objects that appear on the screen**.

- 以不强迫用户进行不必要或不希望的操作的方式来定义交互模式
- 提供灵活的交互方式
- 允许用户交互可中断和可撤消
- 随着技能水平的提高，简化交互并允许定制交互
- 对用户隐藏内部技术
- 设计与屏幕上出现的对象直接交互

Reduce the User's Memory Load

- ❖ Reduce demand on short-term memory.
- ❖ Establish meaningful defaults.
- ❖ Define shortcuts that are intuitive.
- ❖ The visual layout of the interface should be based on a real world metaphor.
- ❖ Disclose information in a progressive fashion.

- 减少对短期记忆的需求
- 建立有意义的默认值
- 定义直观的快捷方式
- 界面的视觉布局应该基于真实世界
- 以渐进的方式披露信息

Make the Interface Consistent

- ❖ **Allow the user to put the current task into a meaningful context.**
- ❖ **Maintain consistency across a family of applications.**
- ❖ **If past interactive models have created user expectations, do not make changes unless there is a compelling reason to do so.**

- 允许用户将当前任务放入有意义的上下文中（明确当前的界面是做什么的）
- 在一系列产品中保持一致性
- 如果过去的交互式模型已经创造了用户期望，除非有令人信服的理由，否则不要进行更改

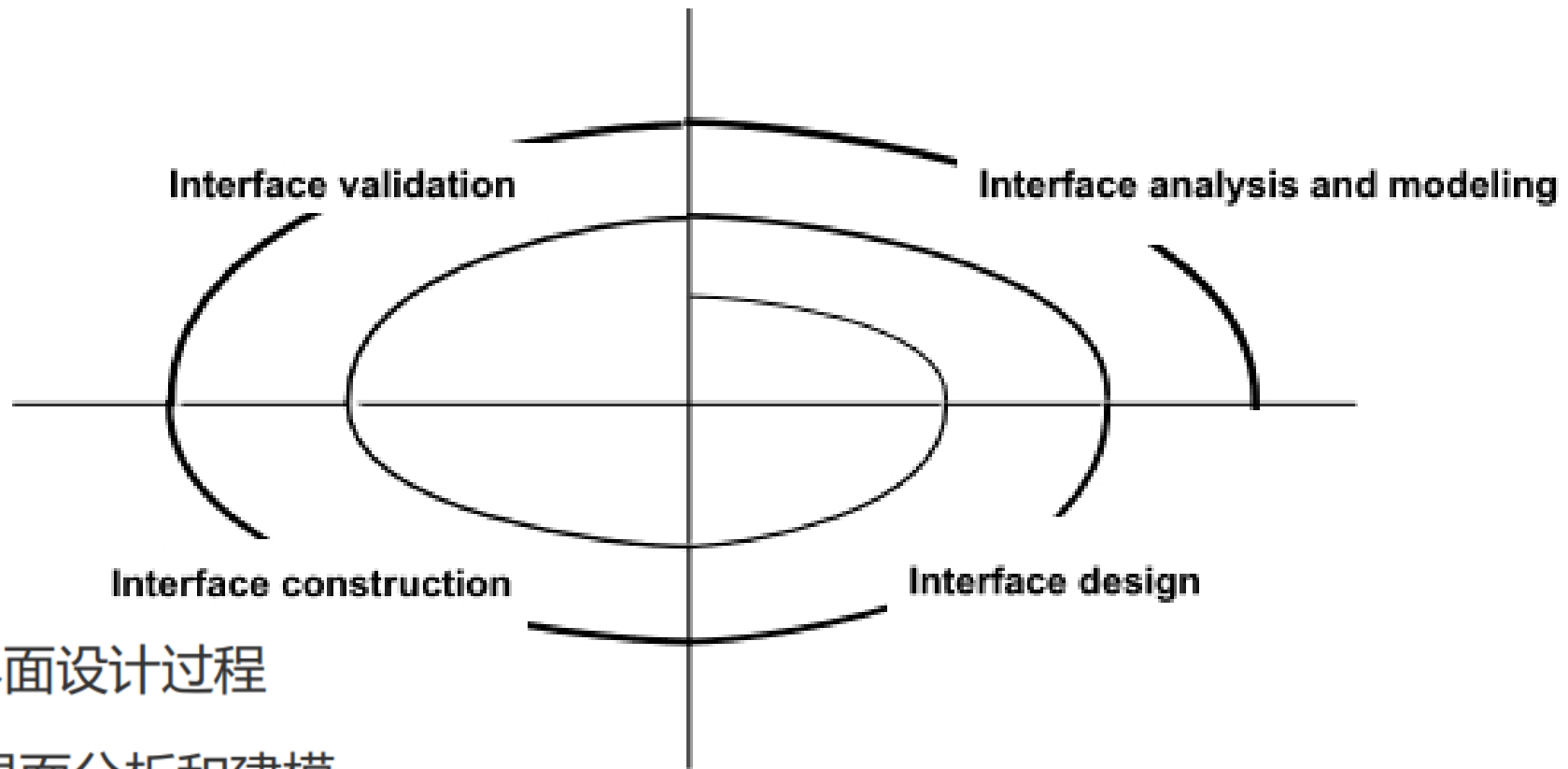
用户界面设计模型

User Interface Design Models

- ❖ **User model** — a profile of all end users of the system
- ❖ **Design model** — a design realization of the user model
- ❖ **Mental model (system perception)** — the user's mental image of what the interface is
- ❖ **Implementation model** — the interface “look and feel” coupled with supporting information that describe interface syntax and semantics

- **用户模型**：所有终端用户的档案
- **设计模型**：用户模型的设计实现
- **心理模型（系统感知）**：用户对界面的心理映像
- **实现模型**：界面的“外观和感觉”，结合描述界面语法和语义的支持信息

User Interface Design Process



2. 用户界面设计过程

- 界面分析和建模
- 界面设计
- 界面构造
- 界面确认

Interface Analysis

❖ Interface analysis means understanding

- (1) **the people (end-users)** who will interact with the system through the interface;
- (2) **the tasks** that end-users must perform to do their work,
- (3) **the content** that is presented as part of the interface
- (4) **the environment** in which these tasks will be conducted.

3. 界面分析

- 最终用户
- 执行的任务（可以使用泳道图）
- 界面的内容
- 被使用的环境（办公室等）

User Analysis

- ❖ Are users trained professionals, technician, clerical, or manufacturing workers?
- ❖ What level of formal education does the average user have?
- ❖ Are the users capable of learning from written materials or have they expressed a desire for classroom training?
- ❖ Are users expert typists or keyboard phobic?
- ❖ What is the age range of the user community?
- ❖ Will the users be represented predominately by one gender?
- ❖ How are users compensated for the work they perform?
- ❖ Do users work normal office hours or do they work until the job is done?
- ❖ Is the software to be an integral part of the work users do or will it be used only occasionally?
- ❖ What is the primary spoken language among users?
- ❖ What are the consequences if a user makes a mistake using the system?
- ❖ Are users experts in the subject matter that is addressed by the system?
- ❖ Do users want to know about the technology the sits behind the interface?

Task Analysis and Modeling

❖ **Answers the following questions ...**

- **What work will the user perform in specific circumstances?**
- **What tasks and subtasks will be performed as the user does the work?**
- **What specific problem domain objects will the user manipulate as work is performed?**
- **What is the sequence of work tasks—the workflow?**
- **What is the hierarchy of tasks?**

Architecture Evaluator - Project Import

Project Import Data Collection Maturity Measurement Evolution Evaluation

Project name :

Project version :

Architecture documents :

Please operate according to following sequence :

Architecture documents import

▼

Data collection

▼

SSAM,DSAM

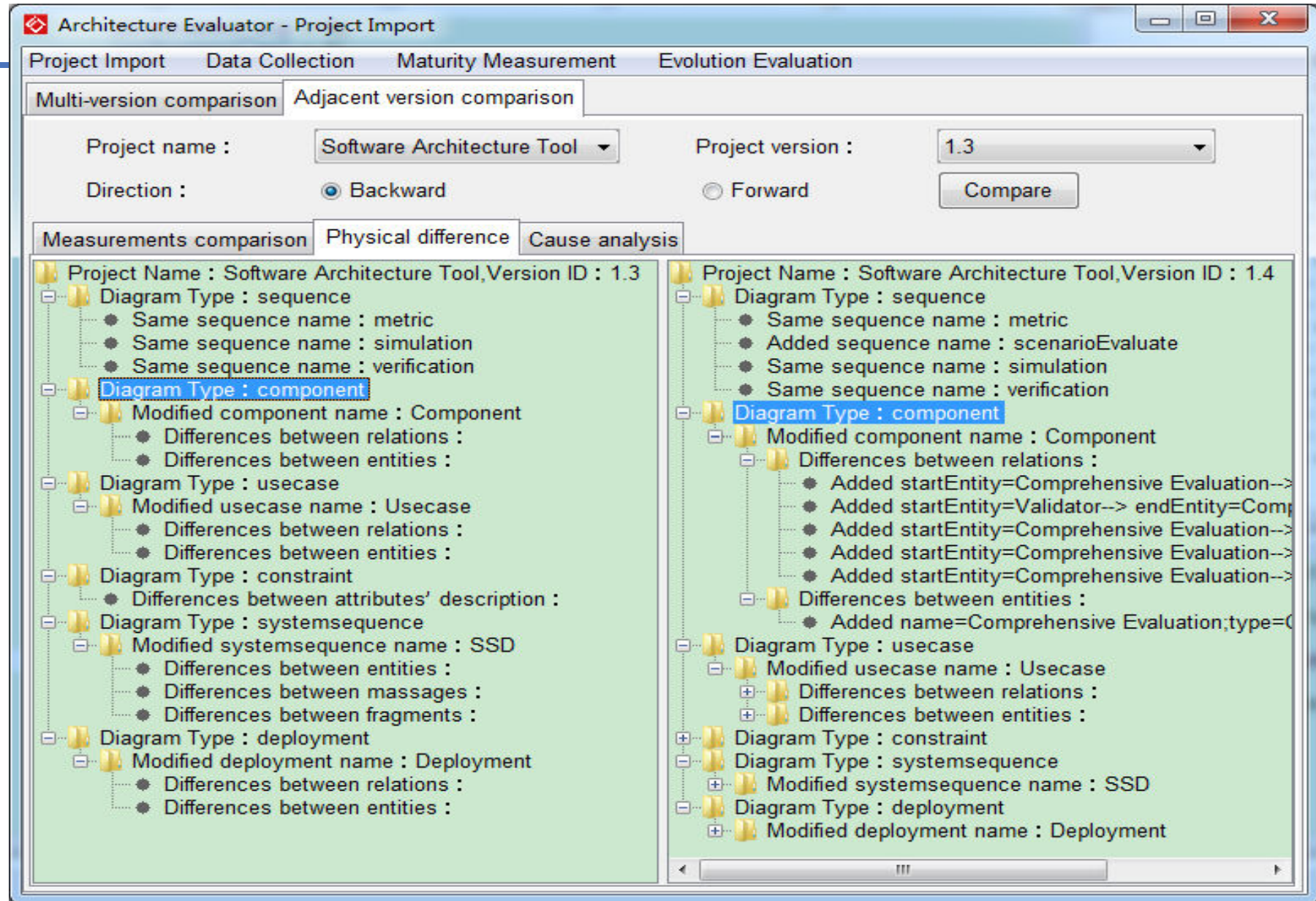
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ISAM

At any time, you can use the Evolution evaluation function for adjacent version evolution.

Analysis of Display Content

- ❖ Are different types of data assigned to consistent geographic locations on the screen (e.g., photos always appear in the upper right hand corner)?
- ❖ Can the user customize the screen location for content?
- ❖ Is proper on-screen identification assigned to all content?
- ❖ If a large report is to be presented, how should it be partitioned for ease of understanding?
- ❖ Will mechanisms be available for moving directly to summary information for large collections of data.
- ❖ Will graphical output be scaled to fit within the bounds of the display device that is used?
- ❖ How will color to be used to enhance understanding?
- ❖ How will error messages and warning be presented to the user?



Interface Design Steps

- ❖ Using information developed during interface analysis, **define interface objects and actions (operations).**
- ❖ **Define events (user actions)** that will cause the state of the user interface to change. Model this behavior.
- ❖ **Depict each interface state** as it will actually look to the end-user.
- ❖ **Indicate how the user interprets the state of the system** from information provided through the interface.

5. 界面设计步骤

- 使用在界面分析过程中开发的信息，定义界面对象和动作（操作）
- 定义将导致用户界面状态改变的事件（用户操作）。为这种行为建模
- 描述每个界面状态
- 说明用户如何根据通过界面提供的信息来解释系统的状态

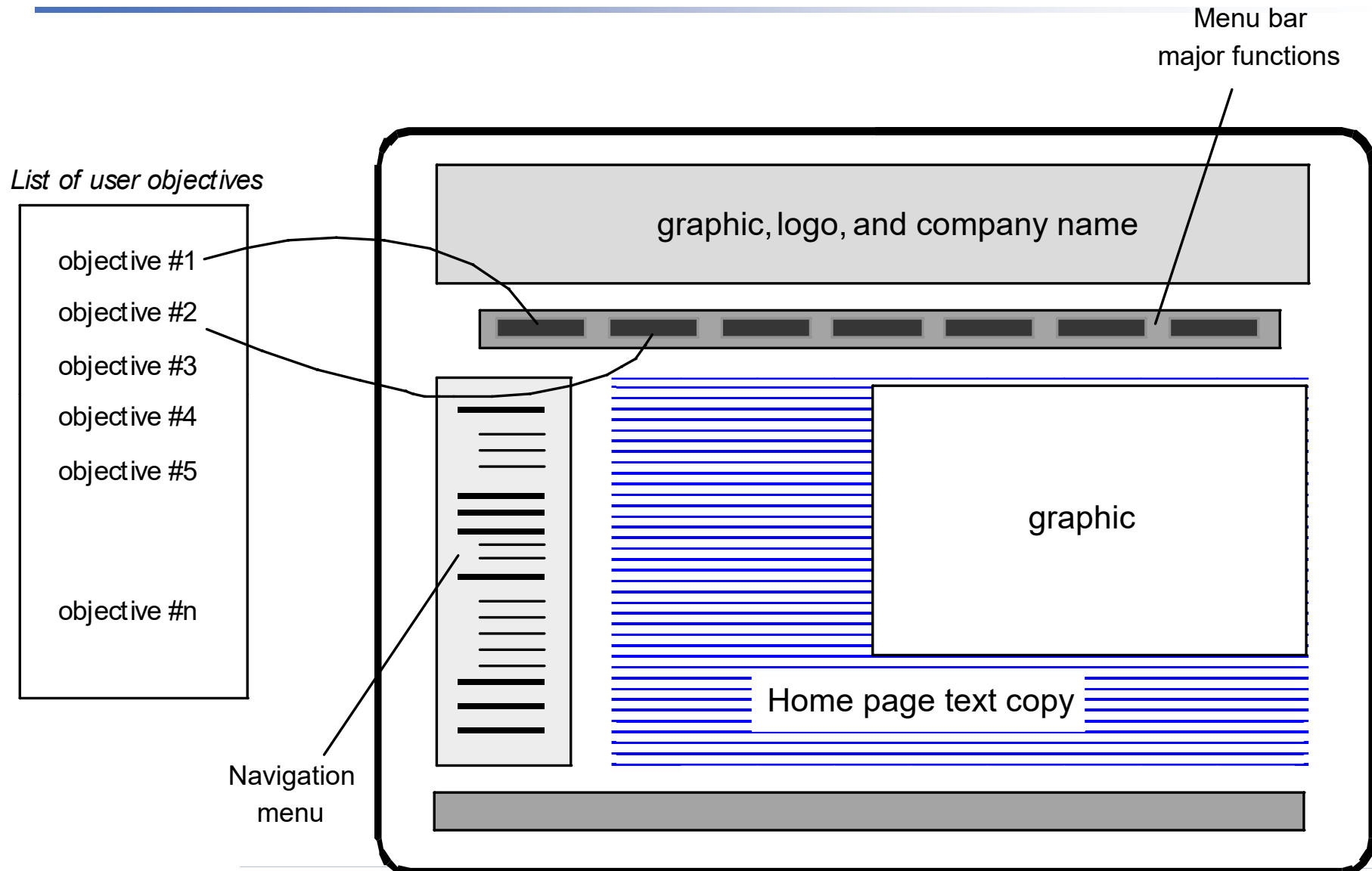
Design Issues

- ❖ **Response time (1s or 0.1s-2.5s?)**
- ❖ **Help facilities**
- ❖ **Error handling**
- ❖ **Menu and command labeling**
- ❖ **Application accessibility**
- ❖ **Internationalization**

6. 界面设计时的问题

- 响应时间
- 帮助
- 错误处理
- 菜单和命令
- 国际化

Mapping User Objectives

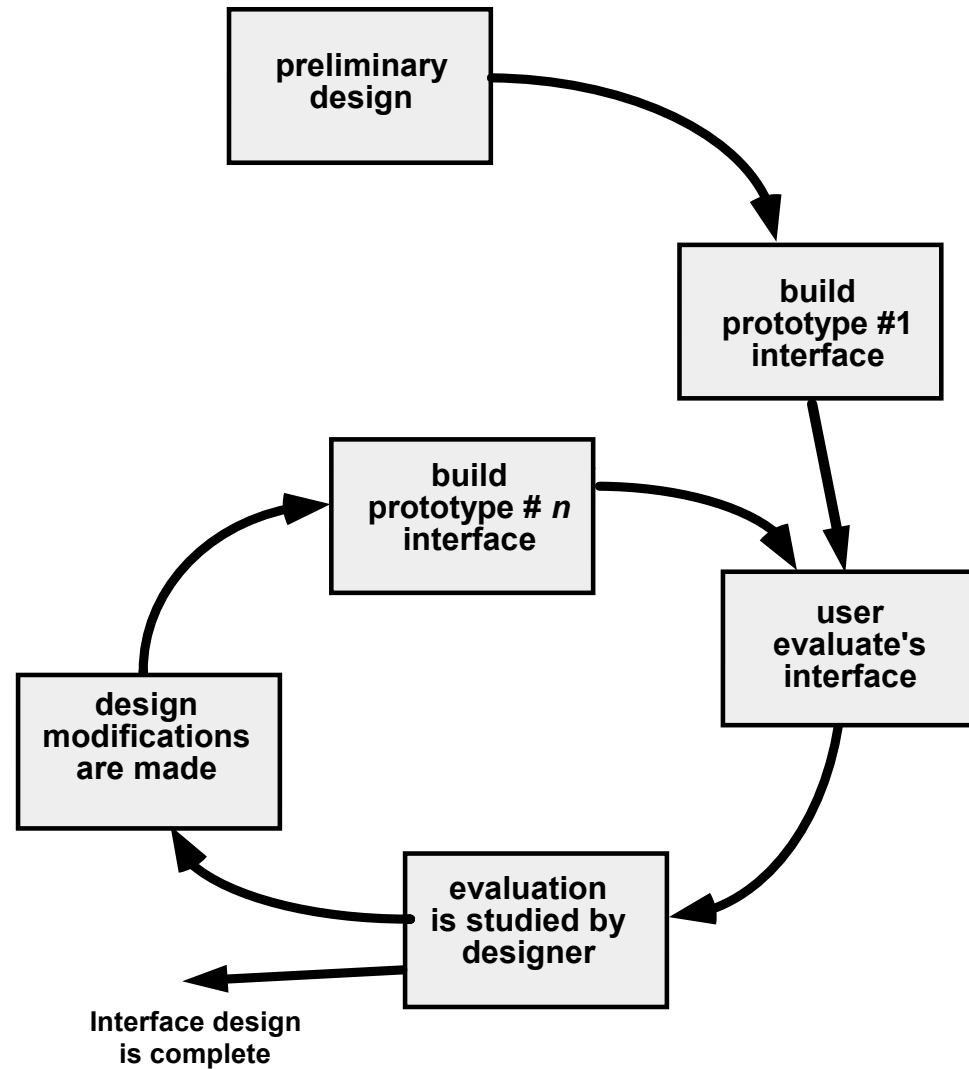


Aesthetic Design （设计美学）

- ❖ **Don't be afraid of white space.**
- ❖ **Emphasize content.**
- ❖ **Organize layout elements from top-left to bottom right.**
- ❖ **Group navigation, content, and function geographically within the page.**
- ❖ **Don't extend your real estate with the scrolling bar.**
- ❖ **Consider resolution and browser window size when designing layout.**

- 不要害怕留白。
- 强调内容。
- 从左上到右下组织布局元素。
- 在页面内地理性地对导航、内容和功能进行分组。
- 不要通过滚动条来扩展你的显示区域。
- 在设计布局时考虑分辨率和浏览器窗口大小。

Design Evaluation Cycle





Southeast University



Thank You !

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